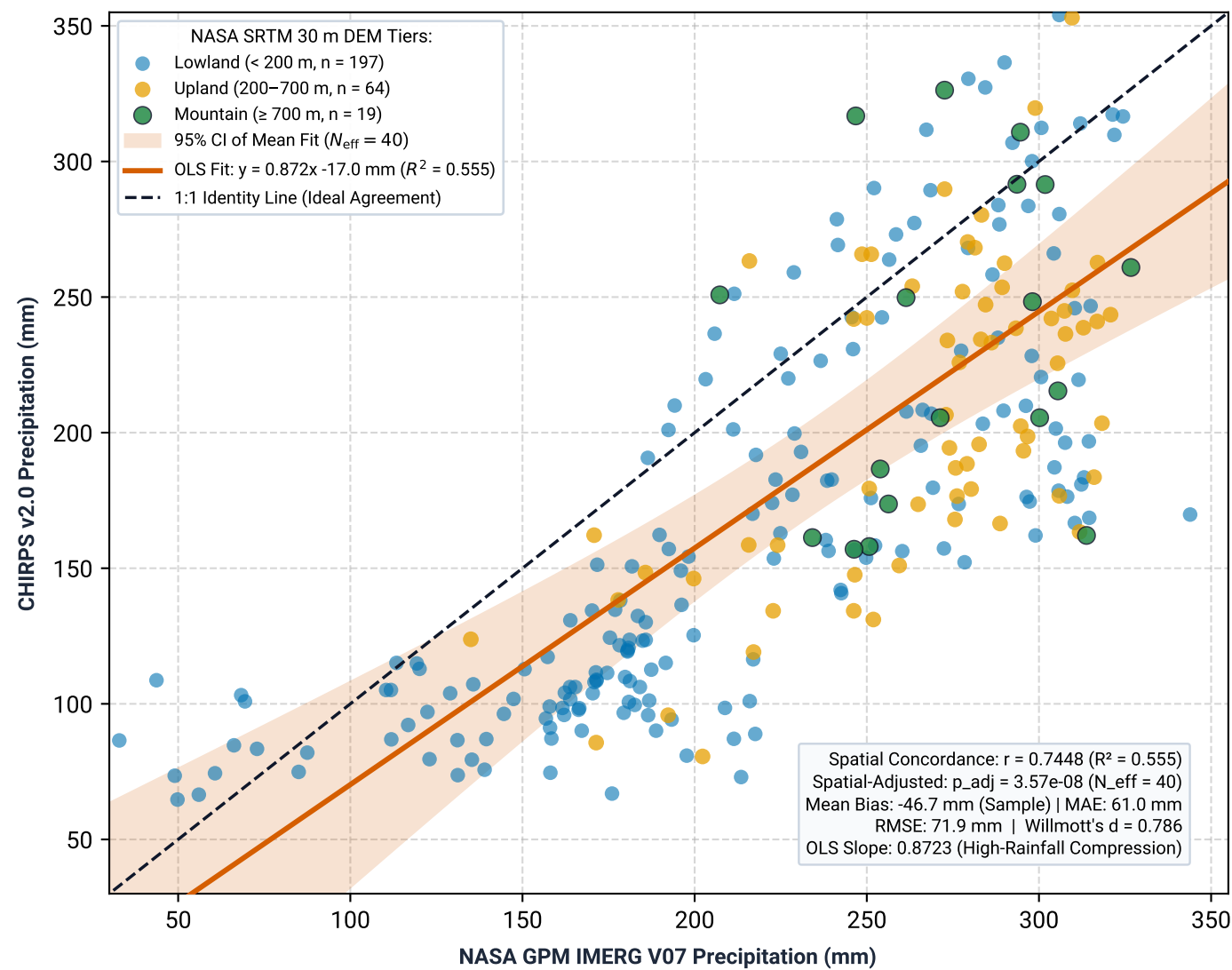
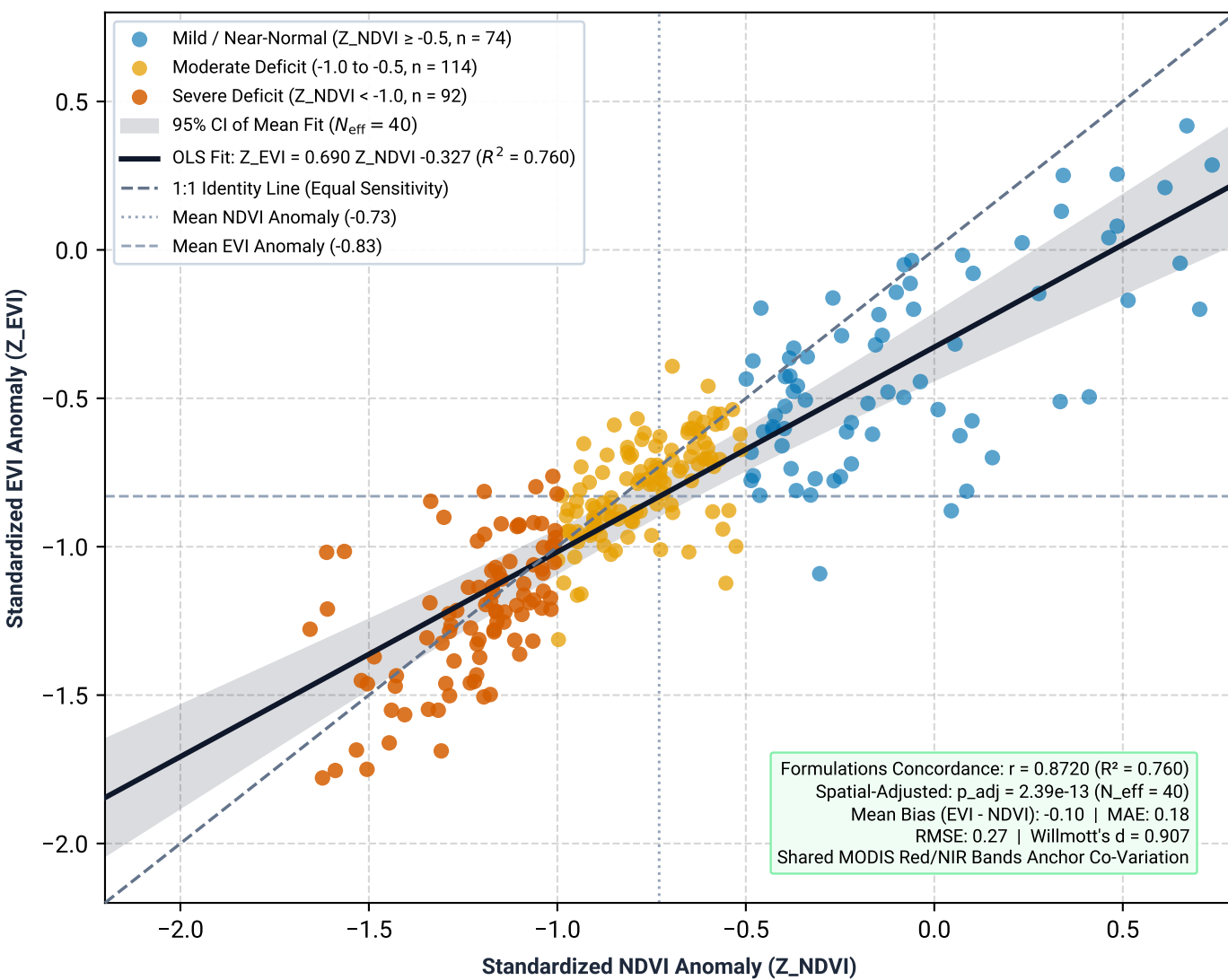


(a) Climate Precipitation Validation: CHIRPS v2.0 vs NASA GPM IMERG V07
Infrared/Gauge vs Radar/Microwave | NASA SRTM 30 m DEM Stratification (Peak SON, 8 Episodes)



(b) Spectral Index Sensitivity & Concordance: MODIS NDVI vs EVI
Cross-Formulation Canopy Deficit Consistency across East Java Vegetated Landscapes (Peak SON)



- Spatial Concordance: CHIRPS v2.0 vs NASA GPM shows solid agreement ($r = 0.7448$, $R^2 = 0.555$).
- High-End Compression: OLS slope ($0.8723 < 1$) confirms CHIRPS under-estimates severe rain peaks.
- DEM Stratification: NASA SRTM 30 m dry bias: Lowland -43.0 mm, Upland -60.9 mm, Mountain -37.0 mm ($d = 0.786$).

- Spectral Concordance: MODIS NDVI vs EVI shows robust co-variation ($r = 0.8720$, $R^2 = 0.760$, $d = 0.907$).
- Synchronous Browning: Both indices confirm severe drought (mean $Z_NDVI = -0.73$, $Z_EVI = -0.83$).
- Structural Note: Shared MODIS Red/NIR bands anchor correlation; EVI introduces -0.10 mean shift.